DEEP LEARNING TECHNIQUES FOR FRAUD DETECTION

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Abstract

In most of the paper reviewed, data pipeline for fraud is often enhanced with sampling methods, introducing synthetic data. Additionally, architectural comparison is often evaluated under a non-unified preprocessing pipeline, which makes it difficult to isolate model-specific contributions. This study compares ensemble, transformer and graph-based models under a unified pipeline. Stratified sampling is employed without providing synthetic data to the model, and consistent data preprocessing steps are sustained across all modeling methods. Multi-seed experimental steps are conducted to validate robustness and stability **(Henderson et al., 2018)**. Despite of business-oriented ranking evaluations- including cumulative gain and lift curves- are plotted to reflect practical fraud investigation constraints **(Provost, F., & Fawcett, T., 2001)**. The proposed framework achieves almost 95% fraud capture confidence when assessing the top 40% highest-risk records, corresponding to a lift of 2.2x compared to random ranking. These results highlight the effectiveness of a unified and deployment-aligned evaluation framework for fraud detection.

1. Introduction

Financial fraud detection has become a critical challenge for modern financial institutions due to the increasing scale, speed, and diversity of digital transactions. Effective fraud detection systems must achieve high recall for fraudulent activities while maintaining low false-positive rates in order to minimize customer disruption and operational costs. Tree-based ensemble methods, such as Random Forest and gradient boosting algorithms, have long been central to fraud detection tasks because of their robustness on heterogeneous tabular data and their strong empirical performance in high-dimensional settings

Recent advances in deep learning, particularly transformer-based architectures, have introduced attention-based representation learning mechanisms to tabular modelling. The tabular transformer and the graph neural network (GNN) are intended to model intricate interactions and relationships, which could be

beneficial for fraud detection. Nevertheless, there are only a few studies that compare classical ensemble learning techniques with transformer models on large financial fraud datasets **(Gorishniy et al., 2021)**.

The goal of this research is to perform a controlled comparison between three modeling paradigms: (1) a traditional ensemble-based model (Random Forest) **(Breiman, 2001)**, (2) a transformer-based model for tabular data (Gated Tab Transformer % FT-Transformer) **(Huang et al., 2020; Gorishniy et al., 2021)**, and (3) a graph-based transformer model (Heterogeneous Graph Transformer) along with an encoder-based anomaly detection model **(Hamilton et al., 2017; Hu et al)**. The prime evaluation dataset is the IEEE-CIS Fraud Detection dataset, which is a high-dimensional dataset with missing identity features and a highly imbalanced dataset. For the robustness and generalization of the model to structurally different tabular data, additional experiments will be done using the Bank Marketing dataset **(IEEE Computational Intelligence Society, 2019; Moro et al., 2014)**.

All models are assessed using a unified framework with a leakage-aware preprocessing step using stratified sampling and multi-seed evaluation for a comprehensive comparison and robustness analysis of the models. Besides using common classification metrics such as ROC AUC and PR-AUC, business-oriented ranking metrics are also included to accommodate real-world constraints for conducting a fraud investigation process.

From the experimental results, it can be observed that Random Forest performs well in the primary fraud detection task, especially metrics such as F1-score, ROC-AUC, PR-AUC, which are highly sensitive to recall performance **(Davis & Goadrich, 2006; Van Rijsbergen, 1979; Fawcett, 2006)**. Although, transformer-based architectures exhibit better precision in certain cases, these enhancements are not always reflected in the overall performance in the fraud capture task under the specific conditions used in the experiment. In contrast, there are partial improvements in the encoder-based anomaly detection model, which indicates the benefits of constrained representation learning in tabular data structures.

The contributions of this study are threefold:

1. A leakage-aware and unified comparison of ensemble, transformer-based, and graph-based models on real-world financial datasets.
2. An evaluation framework that integrates both discrimination metrics and business-oriented ranking analysis.
3. An empirical analysis of architectural behaviour in large-scale tabular fraud detection settings.

2. Literature Review

The IEEE-CIS Fraud Detection dataset, containing 590,540 transactions with approximately 3.5% fraudulent cases, has become one of the most widely used benchmarks for studying class imbalance in financial fraud detection. Much of the recent research on this dataset relies on ensemble-based learning strategies combined with synthetic oversampling techniques to address the severe class imbalance problem.

Several studies demonstrate the strong performance of ensemble-based approaches in this context. For instance, **Moradi et al. (2025)** proposed a preprocessing pipeline that reduces the initial feature space to 167 baseline features and subsequently expands it to 247 features through additional feature engineering. Their framework combines stacking ensembles with oversampling techniques such as SMOTE, ADASYN, and Borderline-SMOTE, achieving high performance with a ROC-AUC of 91.8% and a PR-AUC of 89.1%. Similarly, **Almalki & Masud (2025)** evaluated stacking ensemble architectures incorporating explainable AI components for financial datasets including IEEE-CIS. Their models also rely on synthetic oversampling strategies to address class imbalance, achieving strong ROC-AUC and accuracy results. However, these studies primarily focus on threshold-based metrics, while ranking-based metrics such as cumulative gain and lift—which are particularly relevant in fraud detection scenarios—are rarely incorporated into the evaluation.

Alongside classical ensemble approaches, transformer-based architectures have recently been proposed for tabular data. The SAINT architecture introduced by **Somepalli et al. (2021)** utilizes both self-attention and intrasample attention mechanisms and demonstrates strong AUROC performance across several tabular classification benchmarks, including the Bank Marketing dataset. However, the absence of highly imbalanced financial datasets such as IEEE-CIS in the SAINT evaluation raises questions about the

effectiveness of transformer-based tabular models under conditions of extreme class imbalance and high-dimensional feature spaces.

More broadly, several empirical studies suggest that deep learning architectures do not consistently outperform optimized ensemble-based methods in structured tabular environments. **Borisov et al. (2021)** highlight that traditional tree-based and boosting models frequently dominate deep learning models when extensive feature engineering is present. Similarly, **Gorishniy et al. (2021)** demonstrate that although neural architectures can perform competitively, they are not guaranteed to surpass well-tuned ensemble models across different tabular datasets. These findings suggest that increasing model complexity does not necessarily translate into improved performance for structured tabular data.

Another line of research explores the use of graph-based learning methods to incorporate relational inductive biases. Graph neural networks (GNNs) can capture interactions between entities through neighbourhood aggregation mechanisms **(Hamilton et al., 2017)**, while heterogeneous graph transformers (HGT) extend this capability using attention mechanisms to model complex relational structures **(Hu et al., 2020)**. However, the effectiveness of graph-based models depends heavily on the availability and quality of relational signals beyond what is already encoded in tabular features.

Taken together, existing research highlights several important gaps. Many studies evaluate only a single modelling paradigm—either classical ensembles, deep tabular architectures, or graph-based approaches—making direct comparison difficult. Furthermore, the widespread use of synthetic oversampling may introduce artificial patterns that do not reflect real-world fraud distributions. Finally, evaluation protocols often emphasize threshold-based metrics, while ranking-based metrics that better reflect operational fraud detection scenarios remain underexplored.

Motivated by these limitations, the present study conducts a unified comparison of ensemble, transformer-based, and graph-based architectures using the IEEE-CIS Fraud Detection dataset as the

primary benchmark. In contrast to prior work that heavily relies on synthetic oversampling, this study employs stratified sampling and leakage-aware preprocessing to maintain realistic data distributions. In addition to ROC-AUC and PR-AUC, cumulative gain and lift metrics are incorporated to evaluate ranking efficiency, providing a more comprehensive assessment of model effectiveness in real-world fraud detection environments. To examine cross-domain robustness, the Bank Marketing dataset is also used as a secondary dataset for evaluating model generalization across different tabular classification tasks.

3. Problem Formulation and Hypothesis

Financial fraud detection is inherently characterized by extreme class imbalance, evolving fraud patterns, and strict operational constraints. In practical deployment settings, models must prioritize the ranking of suspicious transactions while maintaining stability under changing data distributions. However, many existing studies address the imbalance problem using synthetic oversampling or data generation techniques, which may alter the natural distribution of transactions and complicate the interpretation of model performance.

Recent research also highlights that evaluating the fidelity, diversity, and generalization properties of synthetic data remains an open challenge, particularly in sensitive domains where data authenticity is critical **(Alaa et al., 2022)**. Artificially generated samples may introduce distributional artifacts that do not accurately reflect real transactional behaviour, raising concerns about whether observed performance gains stem from genuine predictive capability or from artifacts introduced during data generation. For this reason, maintaining a leakage-aware and realistic data pipeline is especially important in financial fraud detection tasks.

In addition, comparative studies across modelling paradigms often lack unified preprocessing frameworks, consistent evaluation metrics, and robustness validation across multiple training runs. Model performance is frequently reported using single-run results or ROC-AUC alone, despite the fact that precision–recall metrics provide a more informative assessment under severe class imbalance. Without controlled experimental design and statistical validation, claims regarding architectural superiority remain difficult to generalize.

To address these limitations, this study implements a unified and leakage-aware experimental pipeline without relying on synthetic data generation. Class imbalance is handled using stratified undersampling applied

exclusively to the training set, thereby preserving the natural distribution of validation and test data. Model performance is evaluated using both PR-AUC and ROC-AUC metrics, while robustness is assessed through multiple random seed experiments and statistical hypothesis testing. Through this controlled experimental framework, the study provides an unbiased comparison of traditional ensemble-based models, transformer-based tabular architectures, and graph neural networks for financial fraud detection.

3.1. Research Questions

1. How do classical ensemble-based models, transformer-based tabular architectures, and graph neural networks compare in fraud detection performance when evaluated under a unified leakage-free experimental pipeline?
2. Do transformer-based tabular architectures provide measurable performance improvements over well-established ensemble models for structured fraud detection datasets?
3. Does incorporating relational information through graph neural networks improve fraud detection performance and model robustness compared with non-relational approaches?

4. Datasets

Two publicly available financial datasets are used to evaluate fraud detection models under different structural conditions. The IEEE-CIS Fraud Detection dataset serves as the fundamental benchmark due to its high dimensionality and sparsity **(IEEE Computational Intelligence Society, 2019)**. The Bank Marketing dataset is used as a complementary experiment to assess model behaviour in a lower-dimensional and less sparse setting **(Moro et al., 2014)**.

4.1 IEEE-CIS Fraud Detection Dataset

The IEEE-CIS Fraud Detection dataset contains real-world e-commerce transactions with a binary target variable (*isFraud*). The dataset consists of transaction and identity tables joined via a transaction identifier. Identity information is missing for many transactions, resulting in substantial feature-level missingness. Missing values are retained, as missingness may carry predictive signal. No imputation is performed.

The original dataset is highly imbalanced. To preserve temporal structure and class proportions while reducing computational cost, stratified sampling aligned with chronological order is applied. The resulting sample is a dataframe of 101,766 transactions with a fraud rate of approximately 14.29%. The data is split into 70% training, 15% validation, and 15% testing.

The dataset initially contains over 400 features. Permutation importance is used as a model-agnostic feature selection method, retaining the top 120 features that preserve approximately 90% of cumulative importance. The same reduced feature set is used across all models.

For graph-based modelling, node-level statistical features (e.g., frequency counts and degree-based statistics) are computed from transactional entities such as users, cards, and devices. The identical feature generation pipeline is applied across classical, transformer-based, and graph-based models to ensure fair comparison.

IEEE-CIS Feature Importance Analysis

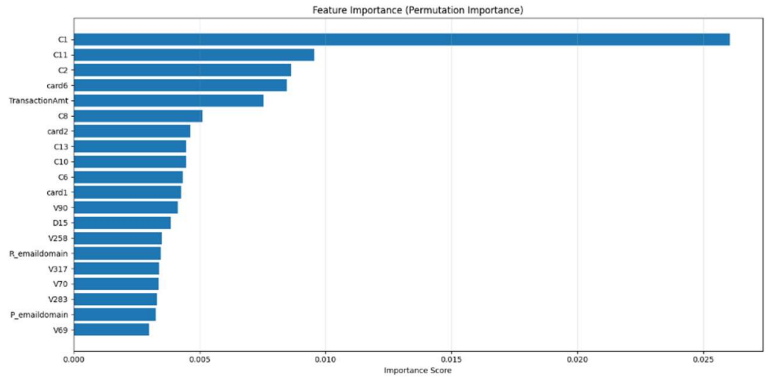

***Figure 3-A**. Permutation-based feature importance analysis of the IEEE-CIS Fraud Detection dataset, illustrating the dominant predictors within the 120-feature reduced representation.*

Permutation importance was computed on a subset, resulting in 120 features out of ~400 total features, with ***Figure 3-A*** showing only the top 20 contributors. The results indicate that fraud prediction is dominated by a small subset of highly influential variables.

The highest contributing feature is Feature C1, followed by Feature C11, Feature C2, card6, and TransactionAmt. Both the anonymous engineered features (denoted by C-series and V-series) and the transactional features have high contributions.

While there are 120 features selected for modelling, the top features represent a significant portion of the predictive influence, implying that the model's performance is concentrated on a small but informative set of features.

4.2 Bank Marketing Dataset

The Bank Marketing dataset from the UCI Machine Learning Repository contains 45,211 instances with a binary target variable indicating whether a client subscribes to a term deposit. The original dataset includes 17 features (including the target variable). After feature engineering, the feature space expands to 50 features. Categorical variables are encoded using one-hot encoding, while continuous variables are standardized. Binary variables and the target column are excluded from the scaling process. To ensure a leakage-free preprocessing pipeline, all preprocessing transformations are fitted on the training data and subsequently applied to the validation and test sets.

The Bank Marketing dataset is included as a secondary benchmark to examine whether model behaviour remains consistent across datasets with different structural characteristics. While the IEEE-CIS dataset contains a large number of engineered features and complex transactional signals, the Bank Marketing dataset represents a comparatively simpler tabular classification problem with fewer features and more stable behavioural patterns. This contrast allows the study to evaluate whether models that perform well on high-dimensional financial transaction data maintain their effectiveness when applied to datasets with lower dimensionality and less complex feature interactions. Evaluating performance across both datasets therefore provides insight into the robustness and generalization ability of the considered architectures.

The dataset has a moderate level of class imbalance, with the positive class comprising 11.7% of the total data points. A training set size of 75% will be used, while the remaining 25% will be for validation and testing. Since the dataset has a manageable number of features, no sampling or reduction is performed.

Node-level frequency and degree features are additionally generated to support graph-based modelling. The same preprocessing pipeline is applied across all models. Node-level frequency and degree features are additionally generated to support graph-based modelling. The same preprocessing pipeline is applied across all models.

4.3 Dataset Characteristics and Implications

Both datasets describe different types of financial prediction environments. The IEEE-CIS dataset is indicative of a big, sparse, and highly encoded fraud detection problem, where there is a lot of missing information, class imbalance, and feature structure, all of which are known to impact the performance of the fraud detection model significantly.

In contrast, the Bank Marketing dataset describes a smaller, more controlled prediction task with a level of imbalance and a small level of sparsity, which allows for a more controlled experiment to better understand model behaviour in a reduced complexity scenario.

The differences between these datasets create a controlled experiment that can be used to better understand how different modelling architectures, through a unified data processing pipeline, address the sparsity, imbalance, and relational aspects of the data, which directly relates to the hypotheses that this study explores.

5. Model Architectures and Experimental Configurations

All models are trained under a unified experimental environment to ensure fair comparison. Each architecture was evaluated across 10 different random seeds to assess robustness to initialization effects.

Rather than applying cross-validation, models were evaluated using repeated controlled experiments (20–25 sequential hyperparameter runs per architecture, over a span of 1 months). Cross-validation was intentionally avoided to prevent methodological imbalance between classical machine learning models and deep learning architectures, which differ substantially in optimization dynamics and computational requirements. Robustness was therefore assessed through repeated fixed-split experimentation.

The evaluated models span three architectural paradigms: a classical ensemble-based method, transformer-based tabular models, and graph neural network (GNN) architectures.

5.1. Random Forest

The traditional baseline model is the Random Forest model, which operates by aggregating a number of decision trees learned under bootstrap aggregation. The model treats imbalance by balancing class weights. The model is computationally efficient to use at test time and provides feature importance, which is interpretable. However, this model does not capture complex feature interactions and relational dependencies between entities.

5.2. Tabular-Transformers

The two Transformer-based models used in the comparison are Gated TabTransformer and FT Transformer. Both models have incorporated attention mechanisms in the models, which enable the models to learn from the complex non-linear relationships between the features.

GatedTabTransformer model incorporates gated embeddings before the attention layers in the model.

FT Transformer employs token feature embeddings, stacked attention blocks, and optimization for tabular data. Training time is longer for Transformer models than for tree-based models.

More flexibility is offered by Transformer models, but interpretability is sacrificed for this flexibility.

Latency is high compared to Random Forest but is acceptable for batch environments.

5.3 Graph Neural Networks

Two graph-based models were implemented: a Heterogeneous Graph Transformer (HGT) and an encoder-only GNN architecture with an anomaly scoring head.
The HGT explicitly models multi-type nodes and relations through attention-based message passing. This enables the capture of relational fraud signals across transactional entities. However, graph construction and heterogeneous attention mechanisms significantly increase computational cost, memory usage, and training time.

The encoder-only GNN model uses a lightweight graph structure while maintaining its ability to learn relations. The model uses a margin-based anomaly model with a dynamic learning rate, which can be tuned using ReduceLROnPlateau to increase training stability **(Ruff et al., 2018)**.

The model is not as computationally expensive as HGT while maintaining graph preprocessing and increased inference latency over tabular models.
Graph-based approaches are the most computationally demanding due to relational aggregation and structural dependency modelling.

5.4 Computational Considerations

From a deployment perspective, computational requirements must be interpreted jointly with architectural complexity and hyperparameter configuration *(see Table 4-A)*. However, the Random Forest model, operating with a total of 500 estimators and a specified tree depth, is still the least computationally intensive model, providing a more efficient inference process because of its non-iterative prediction mechanism. The Transformer-based models, as discussed in detail in *Table 4-A*, incorporate multi-head attention layers and utilize high-dimensional token embeddings, providing a relatively moderate level of computational costs, yet scalable for proper support on

GPUs. On the contrary, graph neural network-based models, including the heterogeneous graph transformer model that incorporates multi-type attention layers, increase the level of computational costs, considering the construction of graph structures, relational message passing, and attention mechanisms. This is also evident when considering runtime costs and architectural details discussed in *Table 4-A*, considering increased memory allocation, dependency on GPUs, and infrastructure costs. Although neural and graph-based models increase the level of representation, they are more computationally expensive, considering a real-world scenario for fraud detection purposes.

Model	Parameters
Random Forest	N_estimators = 500, max_depth = 30, min_samples = 5, class_weight = balanced
GatedTabTransformer	transformer_dim = 128, transformer_depth = 2, transformer_heads = 3, dim_out = 1, batch_size = 128, lr = 3e-3, weight_decay = 3e-4
FT-Transformer	d_token = 128, n_blocks = 2, ffn_d_hidden = 256, attention_dropout = 0.2, ffn_dropout = 0.2, residual_dropout = 0.1, out_dim = 1, batch_size = 1024, lr = 1e-3, weight_decay = 1e-4
HGT	In_dim = 128, h_hid = 192, n_heads = 2, n_layers = 2, dropout = 0.3, binaryNodeHead – in_dim = 192, dropout = 0.3, lr = 2e-2, weight_decay = 1e-4
Encoder-Only GNN	in_dim = 128, hidden_dim = 128, out_dim = 64, alpha = 0.65, dropout = 0.1, num_gnn_layers = 3, num_heads = 2, lr = 3e-4, weight_decay = 2e-4, scheduler = ReduceLROnPlateau(factor = 0.5), anomaly_energy = 0.02, margin = 1.0

Table 5-A. *Hyperparameter configuration of evaluated models across the IEEE-CIS and Bank Marketing datasets.*

6. Methodology

Data Preprocessing Pipeline with two separate branches for two different tasks

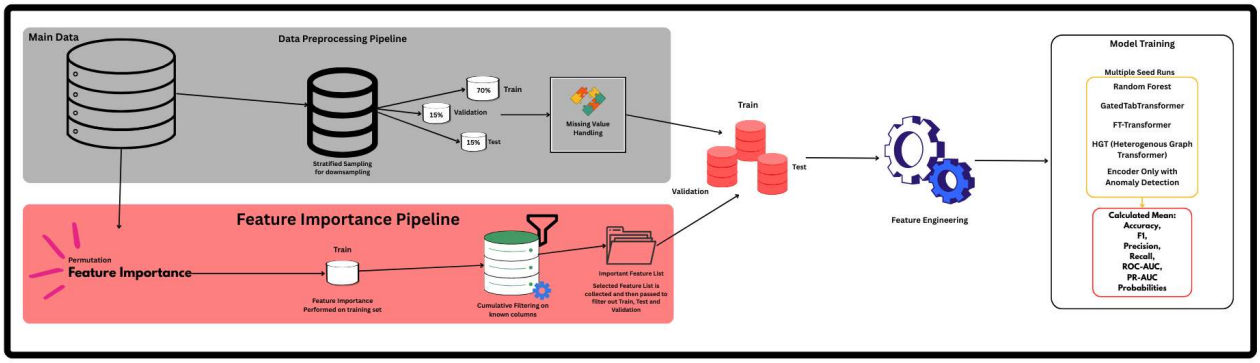

Figure 6-A*. T-Safe Pipeline with 100% purity*

6.1. Leakage-Aware Experimental Pipeline

A unified leakage-aware experimental pipeline was implemented to ensure fair comparison across all model families. First, the raw transaction tables were merged using a unique transaction identifier to create a single analytical dataset. To improve computational efficiency without generating synthetic observations, a fixed random subset of the population was selected within each experimental seed. Transactions were then ordered chronologically using TransactionDT and split into training, validation, and test sets in temporal order to preserve real-world deployment conditions and prevent time leakage.

All preprocessing steps were fitted exclusively on the training data and then applied unchanged to the validation and test sets. Missing values were handled using train-derived imputation statistics, after which permutation importance was used to rank features and retain the most informative subset based on a cumulative contribution threshold. The resulting feature set was applied consistently across all data splits. For graph-based experiments, additional node-level structural features, including degree and normalized degree centrality, were appended to the tabular representation. Continuous variables were then standardized using StandardScaler fitted on the training set only **(Pedregosa et al., 2011)**.

To ensure robustness, the full experimental process was repeated across 10 random seeds, with performance summarized using the mean and standard deviation of the evaluation metrics. Both probabilistic outputs and threshold-based metrics were retained for downstream comparative and statistical analysis. Detailed step-by-step implementation of the pipeline is provided in the Appendix**.**

6.2 Seed-Based Experimental Design and Stability

In order to ensure the robustness of the models, as well as eliminate the variations in the performance of the models due to random initialization, the models were trained and tested over 10 fixed random seeds. This is particularly important in imbalanced fraud detection, where the learning of the minority class may be affected by the random optimization of the model.

PR-AUC was selected as the primary evaluation metric due to its suitability for highly imbalanced classification. For each model, PR-AUC scores were computed per seed and subsequently averaged to obtain the final reported performance. This aggregation strategy helps in reduction of the influence of incidental fluctuations and provides a stable estimate of model behaviour.

By maintaining an identical setting, a fair comparison can be made. The comparison of the model's performance will not be based on the settings but on the model's capabilities, rather randomness.

6.3 IEEE CIS Performance Evaluation

For the performance evaluation of the model on the IEEE-CIS data set, the PR-AUC, cumulative gain, and lift metrics will be used because of sever class imbalance in the data.

IEEE Evaluation Metric (Averaged Metrices Across 10 Seeds)

Model	μ ROC-AUC	μ PR-AUC	μ Accuracy	μ Precision	μ Recall	μ F1-score	μ Specificity
Random Forest	**0.8981**	**0.7512**	**0.8894**	0.6929	0.6327	0.6613	0.9422
HGT	0.8711	0.6813	0.8439	0.5355	0.6752	0.5964	0.8787
Gated	0.8226	0.6429	0.8762	0.7398	0.4265	0.5398	0.9687
FT-Transformer	0.8664	0.6988	0.8195	0.4857	0.7455	0.5860	0.8347
Encoder Only	0.8796	0.7084	0.8653	0.5983	0.6453	0.6205	0.9106

***Table 6.3-A**: Model Metrics on IEEE CIS Fraud Detection Dataset*

Random Forest consistently maintains higher precision across most recall levels, particularly within the mid-recall region that reflects realistic operational thresholds *(Table 6.3-A)*. In fraud detection systems, this region corresponds to the subset of transactions that can feasibly be reviewed by human investigators, where higher precision directly reduces false-positive investigations and associated operational costs *(Table 6.3-A)*. This behaviour is reflected in its highest PR-AUC (≈ 0.75), indicating superior ranking of fraudulent transactions under severe class imbalance. his is further validated by a Complementary Analysis performed on ROC-AUC, which stands at approximately 0.90, signifying a high level of global class separability, hence reinforcing its robustness in discrimination ability. Encoder-Only and FT-Transformer have shown comparable performance with moderately high PR-AUC (≈ 0.71 and ≈ 0.70) and ROC-AUC (≈ 0.88 and ≈ 0.87) values, ensuring the stability of precision retention at increasing recall levels. On the contrary, HGT and GatedTabTransformer have shown comparatively lower PR-AUC, thus emphasizing the lower effectiveness of the model in precision retention at increasing recall levels. The overall performance hierarchy of the models, as indicated by the use of PR-AUC and ROC-AUC, shows the dominance of the IEEE-CIS dataset, where the Random Forest model is the best in providing the most reliable ranking performance in the presence of extreme class imbalance, an important criterion in the context of real-world fraud detection scenarios.

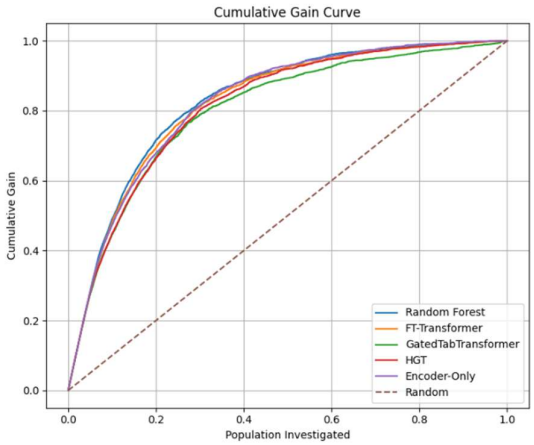

***Figure 6.3-B**: Cumulative Gain Curve – IEEE CIS Dataset*

The cumulative gain curves illustrate the proportion of fraudulent transactions captured as the population is progressively investigated based on model rankings. In fraud detection scenarios, this metric is particularly important because investigators typically review only a limited fraction of transactions flagged as suspicious. As shown in *Figure 6.3-B*, the curves rise steeply in the early portion of the population, indicating that a significant share of fraudulent transactions can be identified within the top 20–30% of investigated cases.

Among the evaluated models, Random Forest demonstrates the strongest early gain behaviour, concentrating more fraudulent transactions in the highest-ranked segment. Transformer-based models follow a similar pattern but exhibit slightly less steep early gains, indicating weaker prioritization of fraudulent cases. The GatedTabTransformer curve appears comparatively flatter, consistent with its lower PR-AUC performance. Overall, the gain curves reinforce the ranking-based evaluation by illustrating how effectively each model prioritizes fraudulent transactions under realistic investigative constraints.

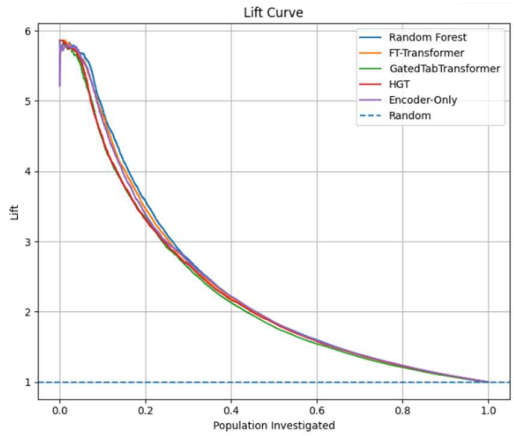

Figure 6.3-C: *Lift Curve – IEEE CIS Dataset*

The lift curves illustrate the degree to which fraudulent transactions are concentrated among the highest-ranked predictions. All models exhibit strong initial lift values, indicating substantial fraud enrichment within the top-ranked population segments. Among them, Random

Forest demonstrates slightly higher lift in the earliest investigated fractions, reflecting a stronger ability to prioritize fraudulent transactions. As the investigated population expands beyond approximately 40–50%, lift values gradually decline toward unity, which is expected as additional non-fraudulent cases are included in the review set.

Taken together with the PR-AUC and cumulative gain results, these patterns suggest that Random Forest not only achieves strong overall discrimination but also provides greater practical value for prioritized fraud investigation. The FT-Transformer and Encoder-Only models form a closely performing group with competitive ranking behaviour, while HGT and GatedTabTransformer exhibit comparatively weaker early ranking efficiency.

6.4 Bank Marketing Validation Analysis

To assess whether the performance hierarchy observed on IEEE-CIS generalizes beyond a single benchmark, models are evaluated on the Bank Marketing dataset using the same three complementary views: PR-AUC, cumulative gain, and lift.

Bank Marketing Evaluation Metric (Averaged Metrices Across 10 Seeds)

Model	μ ROC-AUC	μ PR-AUC	μ Accuracy	μ Precision	μ Recall	μ F1-score	μ Specificity
Random Forest	**0.9250**	**0.5977**	0.8759	0.4812	0.7797	0.5951	0.8886
HGT	0.8916	0.5167	0.8813	0.4944	0.5946	0.5392	0.9192
Gated	0.9082	0.5596	0.7913	0.3481	0.8874	0.4995	0.7786
FT-Transformer	0.9091	0.5500	0.8207	0.3848	0.8543	0.5289	0.8162
Encoder Only	0.9111	0.5739	0.8926	0.5370	0.6407	0.5824	0.9259

***Table 6.4-A**: Model Metrices on Bank Marketing Dataset*

Random Forest remains the best performing model (PR-AUC ≈ 0.60), showing consistent precision-recall trade-offs for the Bank Marketing dataset. Encoder-Only and GatedTabTransformer models closely follow (PR-AUC ≈ 0.57-0.56), while FT-Transformer performs slightly worse (≈0.55), showing weaker ranking performance for positive class instances. When ROC-AUC is added as a

second measure, a similar ranking hierarchy can be observed, with Random Forest showing the highest ROC-AUC (≈0.93) and the neural models (Encoder-Only, FT-Transformer, and GatedTabTransformer) showing similar values at ≈0.91, ≈0.91, and ≈0.91, respectively. This similarity between the results for PR-AUC and ROC-AUC indicates that models that perform better in global discrimination tasks also perform better in precision-recall trade-offs. Nevertheless, the smaller differences in PR-AUC values suggest that improvements in global separability (ROC-AUC) do not transfer to larger improvements in ranking for the minority class. When compared to IEEE-CIS, the differences between models for the Bank Marketing dataset appear smaller, indicating that the dataset, which presents a lower level of imbalance and a less complex structure, may not pose a critical ranking task. The interpretation of the results for both PR-AUC and ROC-AUC shows that the models behave consistently while indicating that ranking sensitivity may not be critical for this validation dataset.

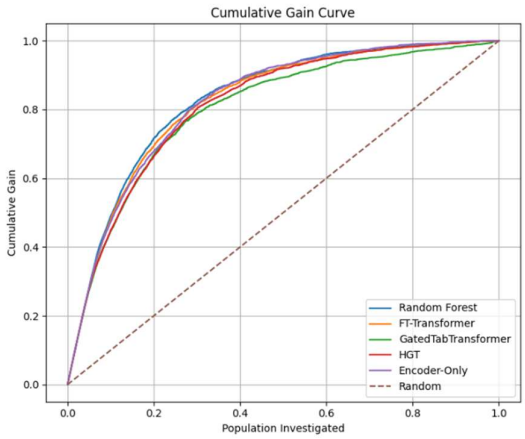

Figure 6.4-B: *Cumulative Gain Curve – Bank Marketing Dataset*

All models significantly outperform the random baseline and demonstrate strong early positive capture within the top 20–30% of the ranked population. This is a demonstration of the early retrieval behavior that is also present in IEEE-CIS, showing that ranking efficiency is preserved across datasets. However, the inter model gap is narrower than in the primary benchmark, implying that dataset characteristics reduce the structural advantage previously held by Random Forest.

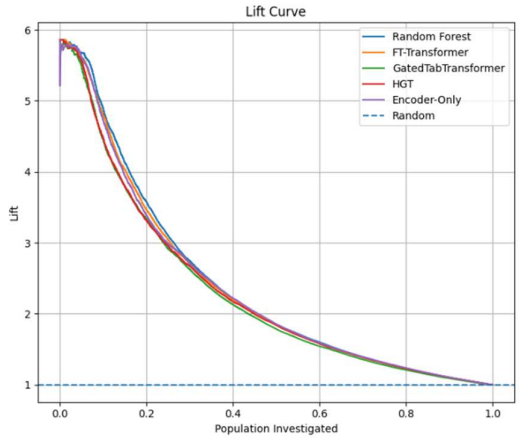

Figure 6.4-C: *Lift Curve – Bank Marketing Dataset*

The lift curves for the Bank Marketing dataset show strong initial lift values across all models, indicating substantial enrichment of positive cases in the highest-ranked segments. Compared with the IEEE-CIS results, the decay of lift is smoother and more similar across models, suggesting that performance differences are less pronounced in this dataset with lower structural complexity and milder imbalance. Consistent with the other evaluation metrics, the HGT model demonstrates comparatively weaker performance, reflected in lower PR-AUC, slower cumulative gain accumulation, and faster lift decay.

Overall, although the magnitude of performance differences is reduced, the general model ordering remains broadly consistent with the IEEE-CIS results. This indicates that the observed ranking behaviour is not specific to a single dataset and that the leading models maintain stable early retrieval efficiency across different tabular prediction settings.

6.5 Effect of Synthetic Oversampling on Model Performance:

Model	ROC-AUC	PR-AUC
Random Forest	**0.796**	**0.353**
HGT	0.761	0.195
Encoder Only	0.790	0.201
GatedTabTransformer	0.508	0.035

FT-Transformer	0.508	0.035

Figure 6.5-A: *Model Performance under SMOTE-Based Oversampling*

To evaluate whether the absence of synthetic oversampling disadvantages neural architectures, an additional sensitivity analysis was conducted using SMOTE within the same leakage-aware experimental pipeline. Importantly, no modifications were made to the original pipeline design; SMOTE was applied directly within the existing workflow and only to the **training data**, while the validation and test sets retained their original distributions. Restricting oversampling to the training set is standard practice in imbalanced learning to prevent information leakage and preserve unbiased evaluation, since applying synthetic sampling before data splitting can introduce artificially generated observations into validation or test sets and lead to overly optimistic estimates of model performance **(Chawla et al., 2002; He & Garcia, 2009)**.

When comparing the results with the original IEEE-CIS experiments, the impact of synthetic oversampling becomes evident. Under the original leakage-aware pipeline, models achieved substantially higher performance, with Random Forest obtaining **PR-AUC = 0.751 and ROC-AUC = 0.898**, while transformer and graph-based models also maintained competitive ranking performance. However, when SMOTE was introduced under the same pipeline conditions, performance deteriorated across several architectures. Transformer-based models such as GatedTabTransformer and FT-Transformer collapsed to near-baseline performance (PR-AUC ≈ 0.035), while graph-based models exhibited noticeably lower ranking performance compared with the original experiments. Although Random Forest retained moderate discrimination ability (PR-AUC ≈ 0.353, ROC-AUC ≈ 0.796), these results remain significantly below the scores obtained under the primary pipeline.

Overall, the comparison indicates that the original leakage-aware pipeline without synthetic oversampling provides more stable and reliable model behaviour for the IEEE-CIS dataset. The

findings suggest that artificially generated samples may distort the natural distribution of financial transactions rather than improving minority-class learning, reinforcing the suitability of the primary experimental design used in this study.

7. Post-hoc Comparative Analysis

The performance of the models was compared by finding the differences in the PR-AUC for each model pair at the same random seeds. This comparison was conducted to ensure a fair comparison while controlling for the variation caused by the seeds. A Wilcoxon signed-rank test was conducted to determine if the differences in the median performance deviated significantly from zero. Moreover, confidence intervals were found for the mean Δ(PR-AUC) to determine the effect. A difference was considered statistically significant when the confidence interval excluded zero and the corresponding p-value was below the predefined significance threshold.

	Model 1	Model 2	Mean Δ(PR-AUC)	CI_low	CI_high	p_value
0	Random Forest	GatedTabTransformer	0.108213	0.099807	0.115428	0.001953
1	Random Forest	FT-Transformer	0.052364	0.047808	0.056815	0.001953
2	Random Forest	HGT	0.069854	0.066500	0.073267	0.001953
3	Random Forest	Encoder-Only	0.042730	0.039355	0.045643	0.001953
4	GatedTabTransformer	FT-Transformer	-0.055850	-0.066722	-0.045457	0.001953
5	GatedTabTransformer	HGT	-0.038359	-0.047633	-0.027718	0.001953
6	GatedTabTransformer	Encoder-Only	-0.065483	-0.072290	-0.057197	0.001953
7	FT-Transformer	HGT	0.017491	0.012750	0.022399	0.001953
9	HGT	Encoder-Only	-0.027124	-0.030555	-0.023203	0.001953
8	FT-Transformer	Encoder-Only	-0.009633	-0.015721	-0.003392	0.027344

Table 7.0-A *Comparative table on Hypothesis Testing for IEEE CIS Dataset*

Across both datasets, paired seed-wise hypothesis testing statistically validates the performance patterns previously observed in the PR-AUC, cumulative gain, and lift analyses. As shown in *Table 7.0-A*, the Random Forest model demonstrates consistently higher PR-AUC than all competing models on the IEEE-CIS dataset. The mean differences in PR-AUC are positive across all pairwise comparisons, including +0.1082 versus GatedTabTransformer, +0.0534 versus FT-Transformer, +0.0699 versus HGT, and +0.0427 versus

Encoder-Only. In each case, the confidence intervals lie entirely above zero and the associated p-values (≈0.001953) indicate statistically significant differences across the seed runs. These results confirm that the superior performance of Random Forest observed in earlier ranking-based evaluations is not attributable to random variation across experimental seeds but reflects a stable and statistically significant advantage.

	Model 1	Model 2	Mean Δ(PR-AUC)	CI_low	CI_high	p_value
0	Random Forest	GatedTabTransformer	0.038084	0.032061	0.044087	0.001953
1	Random Forest	FT-Transformer	0.047645	0.043197	0.052099	0.001953
2	Random Forest	HGT	0.080969	0.077606	0.084529	0.001953
3	Random Forest	Encoder-Only	0.023800	0.019005	0.028626	0.001953
5	GatedTabTransformer	HGT	0.042886	0.037886	0.047542	0.001953
7	FT-Transformer	HGT	0.033324	0.027764	0.039252	0.001953
9	HGT	Encoder-Only	-0.057169	-0.061470	-0.053211	0.001953
6	GatedTabTransformer	Encoder-Only	-0.014283	-0.020371	-0.008218	0.003906
8	FT-Transformer	Encoder-Only	-0.023845	-0.030681	-0.016094	0.003906
4	GatedTabTransformer	FT-Transformer	0.009561	0.002045	0.017753	0.027344

Table 7.0-B: *Comparative table on Hypothesis Testing for Bank Marketing*

A similar phenomenon is observed in the validation benchmark dataset, where, as shown in *Table 7.0-B*, the Δ(PR-AUC) values of the model, Random Forest, are positive and statistically significant over the competing models in the Bank Marketing dataset, such as +0.08097 over HGT, +0.04765 over FT Transformer, +0.03808 over Gated Tab Transformer, and +0.02380 over Encoder Only, where the confidence interval is above zero, although the effect sizes are relatively lower, as expected due to the compressed model separation in the PR curves, although the phenomenon is observed over two structurally different financial prediction tasks. This cross-dataset consistency confirms that the observed superiority is not dataset-specific but robust across structurally different financial prediction tasks

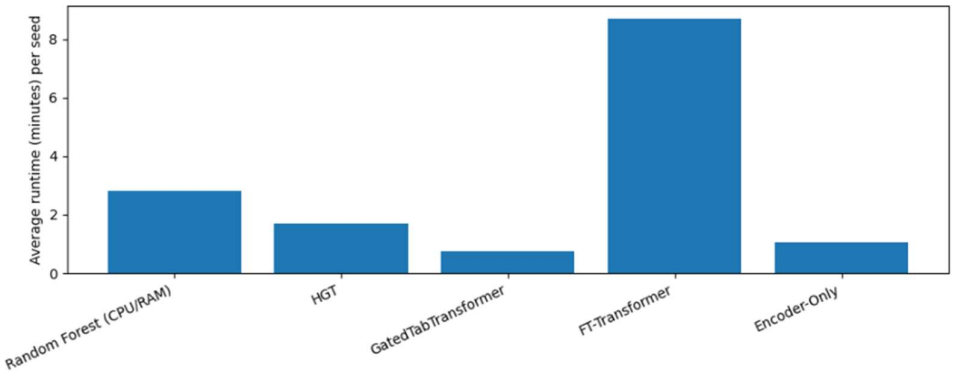

Figure 7.0-A*: Average Runtime per seed*

From an operational perspective, model performance must be considered alongside computational efficiency. *Figure 7.0-A* shows the average runtime per seed, including both training and inference. Random Forest requires approximately **2.8 minutes per seed** on CPU-based high-RAM infrastructure, while the deep learning models rely on GPU resources (H100 for Encoder-Only and GatedTabTransformer, and L4 GPUs for HGT and FT-Transformer). Notably, **FT-Transformer exhibits the highest runtime (≈8.7 minutes per seed)** despite GPU acceleration. Importantly, these additional computational costs do not translate into statistically superior PR-AUC performance compared with Random Forest, as confirmed by the post-hoc comparisons in Tables 7.0-A and 7.0-B.

Taken together, the combination of **higher PR-AUC, statistically significant performance differences, stronger early ranking behaviour (gain and lift), and lower computational cost** suggests that Random Forest offers the most balanced solution in terms of predictive performance, robustness, and operational efficiency. While transformer-based and graph-based models remain competitive, their increased computational requirements do not provide sufficient performance improvements to justify the added complexity in this setting.

8. **Discussion**

This study evaluated three modelling paradigms—classical ensembles (Random Forest), transformer-based tabular architectures (GatedTabTransformer and FT-Transformer), and graph-based models (Encoder-Only and HGT)—across two financial datasets under controlled multi-seed experimentation. The central conclusion is consistent across descriptive evaluation (PR curves, cumulative gain, lift), inferential validation (paired seed-wise testing with confidence intervals and Wilcoxon p-values), and computational profiling: Random Forest provides the most robust and operationally effective performance while remaining the most cost-efficient to deploy.

8.1 Why PR-AUC is the Primary Metric in This Study

Fraud detection is an inherently imbalanced classification problem where the positive class (fraud) is rare and operational decisions depend on prioritizing a small subset of transactions for review. In such settings, ROC-AUC can be overly optimistic because it evaluates the trade-off between true-positive rate and false-positive rate over all thresholds, even when the negative class dominates. Precision–Recall analysis is more directly aligned with the business objective because precision explicitly measures the proportion of investigated cases that are truly fraudulent, and recall quantifies fraud coverage. Prior work has shown that Precision–Recall curves (and PR-AUC) are more informative than ROC curves for imbalanced problems and better reflect performance on the minority class **(Davis & Goadrich, 2006; Saito & Rehmsmeier, 2015)**.

Accordingly, this study uses PR-AUC as the primary metric and complements it with cumulative gain and lift curves, which translate ranking quality into operational value: “how much fraud is captured in the top x% of transactions” and “how enriched that top segment is relative to random selection.” This combined view evaluates not only discrimination but also practical prioritization under limited investigation capacity.

8.2 Random Forest: Robust Superiority Under Seed-Controlled Testing

Across both IEEE-CIS and Bank Marketing, Random Forest achieves the highest mean PR-AUC and statistically significant positive paired Δ(PR-AUC) relative to every other model. On IEEE-CIS, Random Forest exceeds FT-Transformer by approximately +0.053 and Encoder-Only by +0.043, with confidence intervals strictly above zero across comparisons; the Bank Marketing validation benchmark preserves the same directionality with smaller but still significant effect sizes. This demonstrates that Random Forest’s advantage is not an artifact of a single run or a single dataset but persists under seed-controlled pairing and cross-dataset evaluation.

Operationally, the PR-AUC advantage is reinforced by stronger early cumulative gain and higher initial lift concentration. This matters because fraud teams do not investigate the full population; they investigate the highest-scoring segment. Steeper early gain and stronger early lift imply that, for the same review budget, Random Forest surfaces more true fraud cases and wastes fewer investigations on false positives.

A likely explanation is data–model alignment. IEEE-CIS contains heavily engineered and heterogeneous tabular features, where tree ensembles can exploit non-linear interactions and threshold effects directly without requiring representation learning **(Breiman, 2001)**. Prior benchmarking studies consistently show that well-tuned tree-based ensemble models remain highly competitive for structured tabular data and are hard to beat unless dataset scale and/or feature structure are well-suited to representation learning, according to **Borisov et al. (2021)** and **Gorishniy et al. (2021)**.

8.3 Transformer-Based Models: Competitive, but Ranking Efficiency is Less Stable

Both transformer models demonstrate competitive performance in terms of PR-AUC and good early retrieval performance, particularly for IEEE-CIS, suggesting that they are able to learn effective interaction between features. Yet, in all PR curves, there is a degradation in precision at earlier points compared to the Random Forest model, which negatively impacts PR-AUC and operational ranking concentration. Practically, earlier precision decay increases false-positive exposure in the reviewed segment, raising investigation cost and potential customer friction.

Importantly, the paired statistical tests do not provide evidence that either transformer architecture is superior to Random Forest on either dataset. This follows the general tabular DL literature: where attention-based models are known to be potential strong performers but do not

necessarily ensure improvement over the best ensemble methods on a given dataset **(Gorishniy et al., 2021; Somepalli et al., 2021; Borisov et al., 2021)**.

8.4 Graph-Based Modelling: Relational Signal as a Conditional Advantage

The strength of Graph Neural Networks is the use of relational inductive biases through the aggregation of information over connected entities. GraphSAGE proves the effectiveness of aggregating information over the neighbourhood, improving the representation of the nodes with the addition of context **(Hamilton et al., 2017)**. HGT generalizes this approach, by incorporating heterogeneous relation information through the use of attention mechanisms **(Hu et al., 2020)**. This model is effective when the predictive information in the relation cannot be accessed through the attributes of the nodes themselves.

However, the effectiveness of graph modelling depends on whether relational structure provides incremental signal beyond engineered features. In this study, node-level frequency statistics and relational summaries were already incorporated into the tabular representation. Under these conditions, explicit message passing may not introduce additional discriminative power. Empirically, HGT does not outperform tree-based or transformer-based models in PR-AUC or paired statistical testing across either dataset, while incurring greater computational cost. These findings suggest that graph-based architectures offer benefit only when structural information remains underrepresented in tabular features.

8.5 Robustness and Practical Implications

Another important contribution of this study is the use of seed-controlled experimentation in combination with paired non-parametric statistical tests. The results show that Random Forest consistently achieves the strongest performance, with the highest central tendency and the lowest seed-wise dispersion.

From a deployment perspective, computational efficiency further strengthens this conclusion. Random Forest operates on CPU-based high-RAM infrastructure with lower average training time per seed, whereas transformer and graph-based models require GPU acceleration (H100 or L4) and longer runtime without statistically superior gains. When predictive superiority is not accompanied by statistically significant improvement, increased architectural complexity cannot be justified on cost-efficiency grounds.

Overall, the results do not support the assumption that increased architectural sophistication inherently improves fraud detection in structured tabular environments. Instead, they reinforce a growing body of evidence that classical ensemble methods remain highly competitive—and often dominant—when strong feature engineering and tabular structure are present.

9. Methodological Motivation and Research Directions

The empirical results show that classical ensemble methods maintain their effectiveness for structured tabular fraud detection because they use strong feature engineering methods. Meanwhile, transformer models and graph models demonstrate complementary strengths, in recall-oriented detection and relational modelling.

The observed pattern shows that future advancements require researchers to combine different paradigms because both concepts offer valuable benefits. Firstly, tree-based ensembles establish robust inductive bias which benefits heterogeneous tabular features. Future research may explore methods to augment structured feature partitioning or interaction-aware methods within attention-based models to align with tabular data features. Secondly, transformers demonstrate two key abilities which enable them to achieve high recall and develop multiple ways of representing information. Hybrid architectures that combine transformer encoders with tree-based ranking systems will enhance precision and recall performance through their feature abstraction capabilities. Thirdly, the graph-based models provide

relational context which depends on the particular dataset to show its useful effects. Future research should investigate selective relational augmentation which adds graph structure when researchers discover significant relational signals instead of assuming universal advantages from graph modelling. Finally, the models needed consistent preprocessing as well as feature pipelines to ensure fair evaluation results because those two elements protected against bias. Architectural studies should use unified evaluation frameworks to compare their results because preprocessing methods lead to modelling advancement between architectural models. The results indicate that fraud detection development will occur through strategic hybridization and architectural design that matches data characteristics instead of simply increasing model complexity.

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11. Appendix

	seed	Random Forest	GatedTabTransformer	FT-Transformer	HGT	Encoder-Only
0	5	0.752709	0.641784	0.694237	0.686445	0.707788
1	17	0.750789	0.640071	0.696805	0.685570	0.719041
2	32	0.750577	0.637869	0.705415	0.678585	0.704712
3	42	0.750454	0.639854	0.690906	0.685090	0.712244
4	56	0.751062	0.629368	0.709761	0.679850	0.703197
5	73	0.751759	0.655404	0.687330	0.673166	0.705358
6	101	0.751258	0.624792	0.708972	0.689117	0.702987
7	137	0.751992	0.673360	0.701369	0.676944	0.710342
8	2023	0.749332	0.645275	0.696930	0.684215	0.712247
9	9999	0.751646	0.641670	0.696216	0.674054	0.706356

Appendix A: Seed Tabel For IEEE CIS Fraud Detection

	seed	Random Forest	GatedTabTransformer	FT-Transformer	HGT	Encoder-Only
0	5	0.596482	0.561880	0.544482	0.519003	0.579029
1	17	0.598347	0.547810	0.560586	0.507861	0.558235
2	32	0.597495	0.545508	0.546116	0.518504	0.570750
3	42	0.600689	0.551319	0.545972	0.513192	0.571876
4	56	0.597097	0.573547	0.537602	0.521252	0.577695
5	73	0.597015	0.560774	0.557683	0.522274	0.573450
6	101	0.599895	0.569460	0.555453	0.514009	0.571992
7	137	0.596385	0.572148	0.554903	0.523004	0.570984
8	2023	0.596437	0.555396	0.554657	0.517258	0.587794
9	9999	0.596749	0.557912	0.542684	0.510540	0.576780

Appendix B: Seed Table for Bank Marketing

Complete pairwise seed-wise statistical comparisons, are provided in **Appendix C**

Table for IEEE CIS dataset.

- Can help understand how each model is performing; when compared with other models sequentially. This supports how different model is good then the others statistically

	Model 1	Model 2	Mean PR-AUC (M1)	Mean PR-AUC (M2)	Mean Δ(PR-AUC) (M1-M2)	Median Δ(PR-AUC) (M1-M2)	Wilcoxon_stat	p_two_sided	p_one_sided_M1_gt_M2	p_one_sided_M1_lt_M2	Winner (by mean Δ)	Sign	Directional significance
0	Random Forest	GatedTabTransformer	0.751158	0.642945	0.108213	0.110659	0.0	0.001953	0.000977	1.000000	Random Forest	>	Random Forest > GatedTabTransformer
1	Random Forest	HGT	0.751158	0.681304	0.069854	0.068739	0.0	0.001953	0.000977	1.000000	Random Forest	>	Random Forest > HGT
2	Encoder-Only	GatedTabTransformer	0.708427	0.642945	0.065483	0.066908	0.0	0.001953	0.000977	1.000000	Encoder-Only	>	Encoder-Only > GatedTabTransformer
3	FT-Transformer	GatedTabTransformer	0.698794	0.642945	0.055850	0.053500	0.0	0.001953	0.000977	1.000000	FT-Transformer	>	FT-Transformer > GatedTabTransformer
4	Random Forest	FT-Transformer	0.751158	0.698794	0.052364	0.053193	0.0	0.001953	0.000977	1.000000	Random Forest	>	Random Forest > FT-Transformer
5	Random Forest	Encoder-Only	0.751158	0.708427	0.042730	0.045105	0.0	0.001953	0.000977	1.000000	Random Forest	>	Random Forest > Encoder-Only
6	HGT	GatedTabTransformer	0.681304	0.642945	0.038359	0.042688	0.0	0.001953	0.000977	1.000000	HGT	>	HGT > GatedTabTransformer
7	Encoder-Only	HGT	0.708427	0.681304	0.027124	0.027593	0.0	0.001953	0.000977	1.000000	Encoder-Only	>	Encoder-Only > HGT
8	FT-Transformer	HGT	0.698794	0.681304	0.017491	0.017010	0.0	0.001953	0.000977	1.000000	FT-Transformer	>	FT-Transformer > HGT
9	HGT	FT-Transformer	0.681304	0.698794	-0.017491	-0.017010	0.0	0.001953	1.000000	0.000977	FT-Transformer	<	FT-Transformer > HGT
10	HGT	Encoder-Only	0.681304	0.708427	-0.027124	-0.027593	0.0	0.001953	1.000000	0.000977	Encoder-Only	<	Encoder-Only > HGT
11	GatedTabTransformer	HGT	0.642945	0.681304	-0.038359	-0.042688	0.0	0.001953	1.000000	0.000977	HGT	<	HGT > GatedTabTransformer
12	Encoder-Only	Random Forest	0.708427	0.751158	-0.042730	-0.045105	0.0	0.001953	1.000000	0.000977	Random Forest	<	Random Forest > Encoder-Only
13	FT-Transformer	Random Forest	0.698794	0.751158	-0.052364	-0.053193	0.0	0.001953	1.000000	0.000977	Random Forest	<	Random Forest > FT-Transformer
14	GatedTabTransformer	FT-Transformer	0.642945	0.698794	-0.055850	-0.053500	0.0	0.001953	1.000000	0.000977	FT-Transformer	<	FT-Transformer > GatedTabTransformer
15	GatedTabTransformer	Encoder-Only	0.642945	0.708427	-0.065483	-0.066908	0.0	0.001953	1.000000	0.000977	Encoder-Only	<	Encoder-Only > GatedTabTransformer
16	HGT	Random Forest	0.681304	0.751158	-0.069854	-0.068739	0.0	0.001953	1.000000	0.000977	Random Forest	<	Random Forest > HGT
17	GatedTabTransformer	Random Forest	0.642945	0.751158	-0.108213	-0.110659	0.0	0.001953	1.000000	0.000977	Random Forest	<	Random Forest > GatedTabTransformer
18	Encoder-Only	FT-Transformer	0.708427	0.698794	0.009633	0.011846	6.0	0.027344	0.013672	0.990234	Encoder-Only	>	Encoder-Only > FT-Transformer
19	FT-Transformer	Encoder-Only	0.698794	0.708427	-0.009633	-0.011846	6.0	0.027344	0.990234	0.013672	Encoder-Only	<	Encoder-Only > FT-Transformer

Complete pairwise seed-wise statistical comparisons, are provided in **Appendix D**

Table for Bank Marketing Dataset.

- Can help understand how each model is performing; when compared with other models sequentially. This supports how different model is good then the others statistically

	Model 1	Model 2	Mean PR-AUC (M1)	Mean PR-AUC (M2)	Mean Δ(PR-AUC) (M1-M2)	Median Δ(PR-AUC) (M1-M2)	Wilcoxon_stat	p_two_sided	p_one_sided_M1_gt_M2	p_one_sided_M1_lt_M2	Winner (by mean Δ)	Sign	Directional significance
0	Random Forest	HGT	0.597659	0.516690	0.080969	0.079085	0.0	0.001953	0.000977	1.000000	Random Forest	>	Random Forest > HGT
1	Encoder-Only	HGT	0.573859	0.516690	0.057169	0.057214	0.0	0.001953	0.000977	1.000000	Encoder-Only	>	Encoder-Only > HGT
2	Random Forest	FT-Transformer	0.597659	0.550014	0.047645	0.047910	0.0	0.001953	0.000977	1.000000	Random Forest	>	Random Forest > FT-Transformer
3	GatedTabTransformer	HGT	0.559575	0.516690	0.042886	0.041413	0.0	0.001953	0.000977	1.000000	GatedTabTransformer	>	GatedTabTransformer > HGT
4	Random Forest	GatedTabTransformer	0.597659	0.559575	0.038084	0.037539	0.0	0.001953	0.000977	1.000000	Random Forest	>	Random Forest > GatedTabTransformer
5	FT-Transformer	HGT	0.550014	0.516690	0.033324	0.032462	0.0	0.001953	0.000977	1.000000	FT-Transformer	>	FT-Transformer > HGT
6	Random Forest	Encoder-Only	0.597659	0.573859	0.023800	0.024483	0.0	0.001953	0.000977	1.000000	Random Forest	>	Random Forest > Encoder-Only
7	Encoder-Only	Random Forest	0.573859	0.597659	-0.023800	-0.024483	0.0	0.001953	1.000000	0.000977	Random Forest	<	Random Forest > Encoder-Only
8	HGT	FT-Transformer	0.516690	0.550014	-0.033324	-0.032462	0.0	0.001953	1.000000	0.000977	FT-Transformer	<	FT-Transformer > HGT
9	GatedTabTransformer	Random Forest	0.559575	0.597659	-0.038084	-0.037539	0.0	0.001953	1.000000	0.000977	Random Forest	<	Random Forest > GatedTabTransformer
10	HGT	GatedTabTransformer	0.516690	0.559575	-0.042886	-0.041413	0.0	0.001953	1.000000	0.000977	GatedTabTransformer	<	GatedTabTransformer > HGT
11	FT-Transformer	Random Forest	0.550014	0.597659	-0.047645	-0.047910	0.0	0.001953	1.000000	0.000977	Random Forest	<	Random Forest > FT-Transformer
12	HGT	Encoder-Only	0.516690	0.573859	-0.057169	-0.057214	0.0	0.001953	1.000000	0.000977	Encoder-Only	<	Encoder-Only > HGT
13	HGT	Random Forest	0.516690	0.597659	-0.080969	-0.079085	0.0	0.001953	1.000000	0.000977	Random Forest	<	Random Forest > HGT
14	Encoder-Only	FT-Transformer	0.573859	0.550014	0.023845	0.025269	1.0	0.003906	0.001953	0.999023	Encoder-Only	>	Encoder-Only > FT-Transformer
15	Encoder-Only	GatedTabTransformer	0.573859	0.559575	0.014283	0.014913	1.0	0.003906	0.001953	0.999023	Encoder-Only	>	Encoder-Only > GatedTabTransformer
16	GatedTabTransformer	Encoder-Only	0.559575	0.573859	-0.014283	-0.014913	1.0	0.003906	0.999023	0.001953	Encoder-Only	<	Encoder-Only > GatedTabTransformer
17	FT-Transformer	Encoder-Only	0.550014	0.573859	-0.023845	-0.025269	1.0	0.003906	0.999023	0.001953	Encoder-Only	<	Encoder-Only > FT-Transformer
18	GatedTabTransformer	FT-Transformer	0.559575	0.550014	0.009561	0.009677	6.0	0.027344	0.013672	0.990234	GatedTabTransformer	>	GatedTabTransformer > FT-Transformer
19	FT-Transformer	GatedTabTransformer	0.550014	0.559575	-0.009561	-0.009677	6.0	0.027344	0.990234	0.013672	GatedTabTransformer	<	GatedTabTransformer > FT-Transformer

Appendix E. Detailed Experimental Pipeline

The study follows a unified leakage-aware experimental pipeline to ensure fair comparison across all model families while preserving realistic evaluation conditions. The full procedure is described below.

Step 1. Data assembly

The raw transaction tables were merged using a unique transaction identifier to create a single analytical dataset. This ensured that each row corresponded to one unique transaction and that all relevant transactional and identity-level attributes were aligned correctly before preprocessing.

Step 2. Population down sampling for computational efficiency

To reduce computational cost without generating synthetic observations, a subset of the available population was randomly selected prior to model training. This sampling step was applied consistently within each experimental seed and kept fixed across all model comparisons for that seed, ensuring that differences in model performance were attributable to the modelling approach rather than to changes in the sampled population.

Step 3. Temporal data splitting

Transactions were ordered chronologically using **TransactionDT** and split into training, validation, and test sets in temporal order. This preserved the real-world deployment setting in which models are trained on past transactions and evaluated on future transactions. Temporal splitting was used specifically to prevent time-based leakage and to better reflect operational fraud detection conditions.

Step 4. Missing value handling using training-fitted statistics

Missing value imputation was performed using statistics derived only from the training split and then applied unchanged to the validation and test splits. This design prevented leakage of information from future data into the training process. Categorical variables were imputed using the predefined categorical imputation rule implemented in the pipeline, while numerical variables were imputed using train-derived numerical statistics before subsequent scaling and modelling steps.

Step 5. Stable feature selection via permutation importance

Feature importance was estimated using permutation importance on the training data. Importance values were aggregated into a feature importance dataframe, and variables were ranked according to their contribution. A cumulative importance threshold of 90% was then applied to retain the most informative variables while discarding low-utility features. This step reduced feature dimensionality while preserving the majority of predictive information.

Step 6. Consistent column filtering across splits

After feature selection, the training, validation, and test sets were all restricted to the same selected feature subset. This ensured that the same input space was preserved across all splits and across all model families. In practice, this step reduced the feature space substantially, for example from approximately 400 features to around 120 retained variables, depending on the experiment configuration.

Step 7. Graph-based feature engineering

For graph-based experiments, an interaction graph was constructed over the transaction data. Node-level structural statistics were then computed for each transaction, including degree and normalized degree centrality. These graph-derived structural features were appended to the tabular feature matrix to enrich the representation with relational information while maintaining alignment with the existing sample-level dataset.

Step 8. Feature scaling using training-fitted StandardScaler

Continuous variables were standardized using StandardScaler, which was fitted only on the training split and then applied to the validation and test splits using the same learned parameters (Pedregosa et al., 2011). This ensured consistent transformation of the data while preventing leakage from non-training splits. Binary variables and target labels were excluded from scaling.

Step 9. Multi-seed model evaluation

The complete experimental process, including Steps 2 through 8, was repeated across 10 fixed random seeds to evaluate model stability and robustness. For each seed, models were trained on the training split and evaluated on the validation and test splits. Both probabilistic outputs and threshold-based performance metrics were recorded, including PR-AUC, ROC-AUC, and other supporting metrics where applicable. Final performance was reported as the mean and

standard deviation across the 10 runs, and averaged prediction probabilities were retained for downstream comparative analysis and threshold-based reporting.